

Chronological Carry Refinement and p -Adic Root Geometry

A Third-Order Arithmetic Lift of DPDA-Derived Correction Geometry

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Abstract

We introduce a third-order chronological carry refinement whose starting point is the ordered correction geometry extracted from a distinguished cancellation transition in a concrete deterministic pushdown automaton for $L_{2,1} = \{w : N_1(w) = 2N_0(w)\}$. In the nonnegative 2-Dyck sector, the left-state transition $0, 0/\varepsilon$ occurs exactly on down-steps landing at heights $3, 5, 7, \dots$; retaining these events chronologically leads to the nearest-neighbor correction geometry developed in the companion paper. For a prime $p \geq 5$, set $\ell_n = (p+1)n - p$ and define, in the truncated ring $\mathbb{Z}[u]/(u^4)$ after fixing p ,

$$P_{p,n}^{[3]}(u) = (1 + \ell_n u)P_{p,n-1}^{[3]}(u) + p(n-1)\ell_{n-1}u^2P_{p,n-2}^{[3]}(u) - p(n-2)(n-1)u^3P_{p,n-3}^{[3]}(u).$$

The three terms represent a free chronological slot, an adjacent pair interaction, and the first connected triple collision. After normalizing the fixed-column coefficients by falling factorials, we prove that the first three columns are explicit polynomials in the row variable. Modulo p , all interaction terms disappear and the production law degenerates to

$$P_{p,n}^{[3]}(u) \equiv \prod_{j=1}^n (1 + ju) \pmod{p},$$

forcing a universal first-kind Stirling shadow.

The third normalized column is governed by a cubic $F_p(m)$ satisfying

$$F_p(m) \equiv m^2(m-1) \pmod{p}.$$

The simple residue $m \equiv 1 \pmod{p}$ lifts to a unique root $\alpha_p \in \mathbb{Z}_p$, whereas the double residue $m \equiv 0 \pmod{p}$ does not lift even modulo p^2 : it undergoes singular extinction. Over $\mathbb{Q}_p$, the missing pair reappears with valuation $1/2$ and generates the ramified quadratic field

$$\mathbb{Q}_p(\sqrt{-2p}).$$

We obtain a complete valuation law for the scaled third column. The novelty claimed is family-specific: neither carry processes, Stirling numbers, Hessenberg production matrices, nor p -adic lifting are new individually; the contribution is the arithmetic refinement of the DPDA-derived chronological geometry and the resolution forced by its third fixed column.

Keywords. chronological carry refinement; deterministic pushdown automata; correction signatures; Stirling numbers of the first kind; p -adic valuation; Hensel lifting; ramification; Hessenberg matrices.

1 Introduction

The present paper starts from the same concrete deterministic pushdown automaton that motivated the correction-geometry study [2, 3]. The automaton recognizes the counting language

$$L_{2,1} = \{w \in \{0,1\}^* : N_1(w) = 2N_0(w)\},$$

and contains a distinguished left-state cancellation transition $0, 0/\varepsilon$. In the nonnegative 2-Dyck sector this transition occurs exactly when a down-step lands at one of the heights $3, 5, 7, \dots$. Recording those events in chronological order produces the correction-height signatures studied in the companion paper, where their exact minimum realization cost is proved to be nearest-neighbor in chronological order [3].

Thus the discovery route of the present work is

$$\begin{array}{c} \boxed{\text{original DPDA} \longrightarrow \text{ordered cancellations} \longrightarrow \text{correction geometry} \\ \longrightarrow \text{CCR-3} \longrightarrow p\text{-adic geometry}} \end{array}$$

The present paper asks the arithmetic question at the fourth arrow: what structure appears when one retains not only the minimum defect but also low-order connected interactions among consecutive chronological correction slots?

We answer this by defining a third-order chronological carry refinement (CCR-3). The construction is deliberately independent of any previously explored or incompletely recorded carry triangle. Its primary object is the recurrence in Definition 3.1; all results in this paper follow from that definition. The pushdown-derived correction geometry supplies the ordered local skeleton, while the arithmetic weights introduced here form an additional polarization of that skeleton. We do *not* claim that the automaton transition table alone forces the numerical weights below.

The first structural result is fixed-column polynomialization through order three. If

$$C_{n,k}^{(p)} = [u^k]P_{p,n}^{[3]}(u), \quad \Pi_{p,k}(m) = \frac{C_{m-1,k}^{(p)}}{(m-1)_k},$$

then $\Pi_{p,k}$ is a polynomial in m for $k = 1, 2, 3$, with explicit closed forms. The second structural result is a universal congruence: modulo p , every genuine carry interaction vanishes, leaving the elementary symmetric polynomial of $1, \dots, m-1$. Consequently

$$\Pi_{p,k}(m) \equiv \frac{(m-k-1)!}{(m-1)!} \begin{bmatrix} m \\ m-k \end{bmatrix} \pmod{p}, \quad k \leq 3,$$

where $\begin{bmatrix} m \\ r \end{bmatrix}$ is the unsigned Stirling number of the first kind.

The arithmetic geometry first becomes singular in the third column. Its Stirling shadow is

$$S_3(m) = \frac{m^2(m-1)}{48},$$

so modulo p there is one simple residue, $m = 1$, and one double residue, $m = 0$. The exact deformation separates them sharply. The simple residue produces an ordinary Hensel tower. The double residue does not branch in $\mathbb{Z}_p$: every integer $m \equiv 0 \pmod{p}$ has exactly one extra factor of p , so no root survives modulo p^2 . We call this phenomenon *singular extinction*. The two missing algebraic roots nevertheless survive in a ramified quadratic extension and have valuation $1/2$.

Carries and Stirling numbers have well-established connections in the Holte carry matrix and related work [4, 6]; p -adic valuations of Stirling numbers are also a developed subject [5, 1]. Likewise, lower-Hessenberg production matrices occur naturally in weighted lattice-path theory [7]. Our claim is therefore not priority for any of those ingredients. The contribution is the family-specific chain

$$\begin{array}{c} \text{chronological local interactions} \longrightarrow \text{fixed-column polynomials} \longrightarrow \text{Stirling shadow} \\ \longrightarrow \text{simple lifting versus singular extinction and ramification.} \end{array}$$

Main results. For primes $p \geq 5$ we prove:

- (i) a three-band Hessenberg/operator realization of CCR-3 and exact transport equations for its first three normalized columns;
- (ii) explicit formulas for $\Pi_{p,1}, \Pi_{p,2}, \Pi_{p,3}$ and the universal first-kind Stirling shadow modulo p ;
- (iii) a unique Hensel root $\alpha_p \equiv 1 \pmod{p}$, singular extinction in the residue $0 \pmod{p}$, and a complete valuation law for the scaled third column;
- (iv) an exact discriminant calculation showing that the two nonintegral roots generate $\mathbb{Q}_p(\sqrt{-2p})$ and have valuation $1/2$.

2 Source geometry and the scope of the arithmetic lift

2.1 The motivating DPDA

Figure 1 displays the concrete 2:1 deterministic pushdown automaton from which the correction statistic was extracted [2, 3]. The full machine recognizes $L_{2,1}$; for the present paper, the relevant feature is only the heavier left-state loop

$$0, 0/\varepsilon,$$

which consumes an input 0 and removes an exposed stack symbol 0.

Under the path encoding $1 \mapsto +1$ and $0 \mapsto -2$, accepted words in the nonnegative sector become 2-Dyck excursions. The state/stack residual invariant of the machine implies that the distinguished transition occurs if and only if the corresponding down-step lands at

$$3, 5, 7, \dots$$

This correspondence is proved in the companion defect-geometry paper [3]; here it fixes the concrete automata-theoretic origin of the chronological events.

2.2 From cancellation events to chronological locality

The companion correction-geometry paper passes from these DPDA-derived events to generalized Dyck paths and associates to a chronological correction history a normalized signature

$$r = (r_1, \dots, r_m).$$

Its main realization functional has the nearest-neighbor form

$$\Delta(r) = r_m + \sum_{i=1}^{m-1} \psi(r_i - r_{i+1}), \quad \psi(a) = \begin{cases} 0, & a \leq 1, \\ a, & a \geq 2. \end{cases}$$

and equals the minimum path defect needed to realize the prescribed ordered history [3].

The only feature needed here is chronological locality: the intrinsic realization cost couples consecutive correction positions. Thus the correction history carries a natural path graph

$$1 - 2 - \dots - n.$$

This is the precise structural inheritance from the DPDA side: the automaton supplies the distinguished cancellation event, the path correspondence orders those events, and the realization theorem turns the resulting history into a nearest-neighbor geometry. CCR-3 begins only after this point, by assigning arithmetic weights to connected chronological supports. Connected supports of size at most three on a path are necessarily a singleton, an adjacent pair, or a consecutive triple.

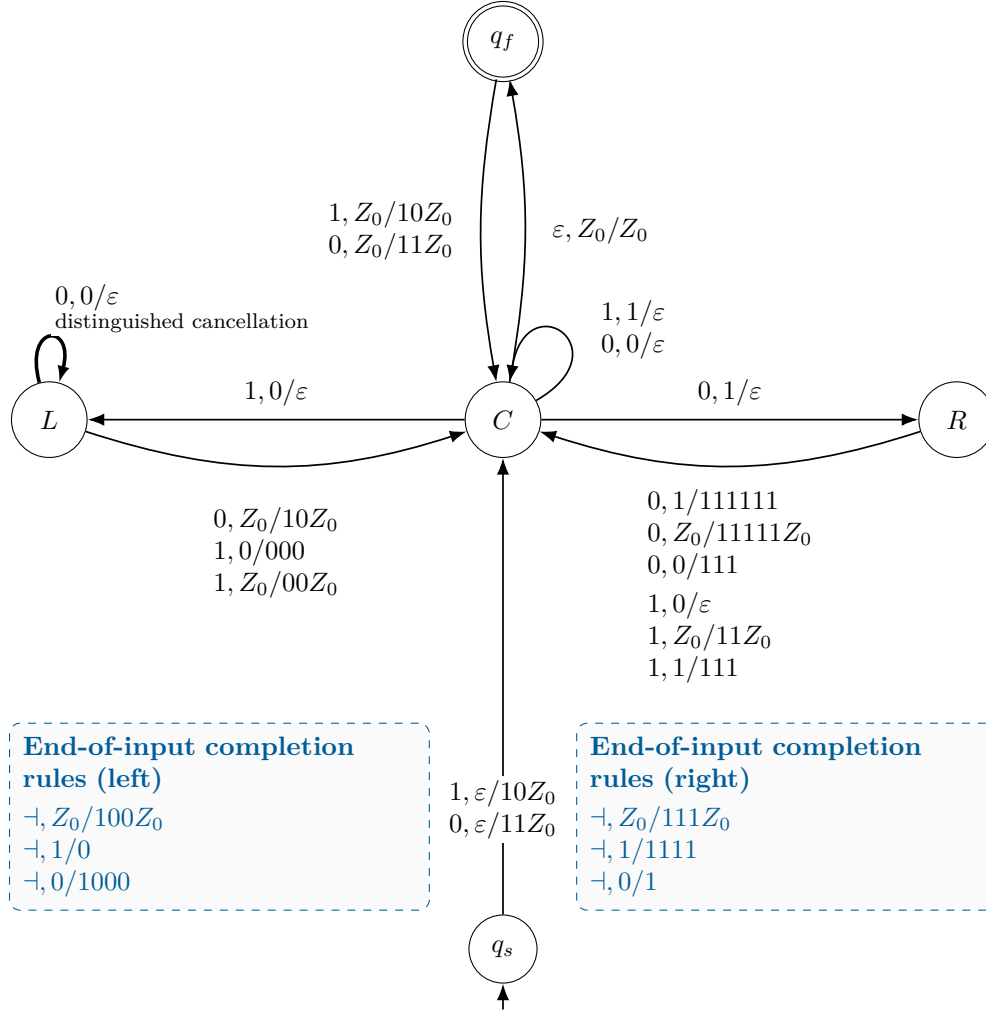


Figure 1: The motivating 2:1 DPDA. The heavier left-state loop is the distinguished cancellation $0, 0/\varepsilon$ from which the correction events are extracted. The dashed right-endmarker completion rules are displayed to preserve the machine's full lineage but play no role in CCR-3.

Lemma 2.1 (Third-order connected support). *Let a chronological interaction be local with respect to the path graph on correction positions. Every connected support of cardinality at most three is one of*

$$\{j\}, \quad \{j, j+1\}, \quad \{j, j+1, j+2\}.$$

Proof. Every connected subset of a path graph is an interval. Intervals of cardinality one, two, and three have exactly the displayed forms. $\square$

Remark 2.2 (Scientific boundary). Lemma 2.1 is the part inherited from the correction geometry. The numerical weights assigned to these supports below are an arithmetic polarization introduced in the present paper. Consequently, the valid implication is

$$\text{pushdown-derived chronological locality} + \text{the specified carry polarization} \implies \text{CCR-3},$$

not the stronger statement that the pushdown automaton by itself determines the arithmetic coefficients.

3 The third-order chronological carry refinement

Fix a prime $p \geq 5$ and put

$$\ell_n = (p+1)n - p. \tag{1}$$

Definition 3.1 (CCR-3). Work in the truncated ring $\mathbb{Z}[u]/(u^4)$ after fixing p . Set

$$P_{p,0}^{[3]}(u) = 1, \quad P_{p,-1}^{[3]}(u) = P_{p,-2}^{[3]}(u) = 0,$$

and, for $n \geq 1$,

$$\boxed{\begin{aligned} P_{p,n}^{[3]}(u) &= (1 + \ell_n u) P_{p,n-1}^{[3]}(u) \\ &\quad + p(n-1) \ell_{n-1} u^2 P_{p,n-2}^{[3]}(u) \\ &\quad - p(n-2)(n-1) u^3 P_{p,n-3}^{[3]}(u) \pmod{u^4}. \end{aligned}} \tag{2}$$

The three nontrivial weights are called respectively the singleton slot, adjacent pair carry, and connected triple collision.

The first rows are

$$\begin{aligned} P_{p,0}^{[3]} &= 1, \\ P_{p,1}^{[3]} &= 1 + u, \\ P_{p,2}^{[3]} &= 1 + (p+3)u + 2(p+1)u^2, \\ P_{p,3}^{[3]} &= 1 + (3p+6)u + (4p^2 + 15p + 11)u^2 + 6(p+1)^2 u^3. \end{aligned}$$

3.1 Hessenberg and operator realizations

Let $H_{p,n}$ be the $n \times n$ leading principal truncation of the infinite lower-Hessenberg matrix with

$$(H_p)_{i,i} = \ell_i, \quad (H_p)_{i,i+1} = 1, \quad (H_p)_{i+1,i} = -pi\ell_i, \quad (H_p)_{i+2,i} = -pi(i+1), \tag{3}$$

and all entries below the second subdiagonal equal to zero.

Proposition 3.2 (Hessenberg realization). *If*

$$D_{p,n}(u) = \det(I_n + uH_{p,n}),$$

then

$$D_{p,n} = (1 + \ell_n u)D_{p,n-1} + p(n-1)\ell_{n-1}u^2 D_{p,n-2} - p(n-2)(n-1)u^3 D_{p,n-3}. \quad (4)$$

Hence

$$P_{p,n}^{[3]}(u) \equiv D_{p,n}(u) \pmod{u^4}.$$

Proof. Expand the determinant by the last row, using the unit superdiagonal. The diagonal term contributes $(1 + \ell_n u)D_{p,n-1}$. Pairing the first subdiagonal entry $-p(n-1)\ell_{n-1}u$ with the superdiagonal entry u contributes $+p(n-1)\ell_{n-1}u^2 D_{p,n-2}$. The second-subdiagonal entry $-p(n-2)(n-1)u$ closes with two consecutive superdiagonal entries and contributes $-p(n-2)(n-1)u^3 D_{p,n-3}$. No other last-row entry is present. $\square$

There is also a compact differential-operator form. On the exponential basis

$$e_i(x) = \frac{x^{i-1}}{(i-1)!}, \quad i \geq 1,$$

let $D = \partial_x$, $Xf = xf$, $N = XD$, and

$$L_p = I + (p+1)N.$$

Then

$$De_i = e_{i-1}, \quad Xe_i = ie_{i+1}, \quad X^2e_i = i(i+1)e_{i+2}, \quad L_pe_i = \ell_i e_i.$$

Corollary 3.3 (Operator form). *The matrix H_p is the matrix, in the basis $(e_i)_{i \geq 1}$, of*

$$\boxed{\mathcal{H}_p = D + L_p - pXL_p - pX^2}. \quad (5)$$

Equivalently,

$$\mathcal{H}_p = [1 + (p+1)x - p(p+1)x^2]\partial_x + 1 - px - px^2.$$

4 Fixed-column transport and polynomialization

Define

$$C_{n,k}^{(p)} = [u^k]P_{p,n}^{[3]}(u), \quad 0 \leq k \leq 3, \quad (6)$$

and, for $m \geq k+1$,

$$\Pi_{p,k}(m) = \frac{C_{m-1,k}^{(p)}}{(m-1)_{\underline{k}}}. \quad (7)$$

Coefficient extraction in (2) gives the following normalized transport laws.

Proposition 4.1 (Transport system). *For the indicated ranges,*

$$m\Pi_{p,1}(m+1) = (m-1)\Pi_{p,1}(m) + \ell_m, \quad (8)$$

$$m\Pi_{p,2}(m+1) = (m-2)\Pi_{p,2}(m) + \ell_m\Pi_{p,1}(m) + p\ell_{m-1}, \quad (9)$$

$$m\Pi_{p,3}(m+1) = (m-3)\Pi_{p,3}(m) + \ell_m\Pi_{p,2}(m) + p\ell_{m-1}\Pi_{p,1}(m-1) - p. \quad (10)$$

Proof. For $k = 1, 2, 3$, extract $[u^k]$ from (2), substitute $n = m$, and divide by the corresponding falling factorial. The factors $(m-1)$, $(m-1)(m-2)$, and $(m-1)(m-2)(m-3)$ cancel exactly to give (8)–(10). $\square$

The last source can be displayed in connected form. Since

$$\Pi_{p,1}(m-1) = \frac{\ell_{m-1} - p}{2},$$

the genuinely new third-order source is

$$\boxed{\mathfrak{T}_{p,3}(m) = p \left[\frac{1}{2} \ell_{m-1} (\ell_{m-1} - p) - 1 \right]}. \quad (11)$$

It separates the propagated pair contribution from one connected collision.

Theorem 4.2 (First three fixed columns). *For $p \geq 5$,*

$$\boxed{\Pi_{p,1}(m) = \frac{(p+1)m - 2p}{2}}, \quad (12)$$

$$\boxed{\Pi_{p,2}(m) = \frac{p+1}{24} \left[3(p+1)m^2 - (5p+1)m - 12p \right]}, \quad (13)$$

and

$$\boxed{\begin{aligned} \Pi_{p,3}(m) = \frac{p+1}{48} & \left[(p+1)^2 m^3 + (p^2 - 1)m^2 \right. \\ & \left. - (34p^2 + 18p)m + 56p^2 - 8p \right]. \end{aligned}} \quad (14)$$

The boundary values are

$$\Pi_{p,1}(2) = 1, \quad \Pi_{p,2}(3) = p+1, \quad \Pi_{p,3}(4) = (p+1)^2.$$

Proof. Equation (8), with $\Pi_{p,1}(2) = 1$, is solved by (12); direct substitution verifies the identity and the initial value. Substituting (12) into (9) gives a first-order recurrence with polynomial source; (13) satisfies it and the boundary value $\Pi_{p,2}(3) = p+1$. Finally, inserting (12) and (13) into (10) yields a cubic polynomial source, and direct substitution verifies (14) together with $\Pi_{p,3}(4) = (p+1)^2$. Uniqueness follows from the first-order transport recurrence in each column. $\square$

5 Universal Stirling shadow

Modulo p , the arithmetic interaction terms in CCR-3 disappear.

Theorem 5.1 (Stirling shadow). *For every $n \geq 0$,*

$$\boxed{P_{p,n}^{[3]}(u) \equiv \prod_{j=1}^n (1 + ju) \pmod{(p, u^4)}}. \quad (15)$$

Consequently, for $k \leq 3$ and $m \geq k+1$,

$$C_{m-1,k}^{(p)} \equiv e_k(1, \dots, m-1) = \left[\begin{matrix} m \\ m-k \end{matrix} \right] \pmod{p}, \quad (16)$$

and

$$\boxed{\Pi_{p,k}(m) \equiv S_k(m) := \frac{(m-k-1)!}{(m-1)!} \left[\begin{matrix} m \\ m-k \end{matrix} \right] \pmod{p}}. \quad (17)$$

In particular,

$$S_1(m) = \frac{m}{2}, \quad S_2(m) = \frac{m(3m-1)}{24}, \quad S_3(m) = \frac{m^2(m-1)}{48}. \quad (18)$$

Proof. Since $\ell_n \equiv n \pmod{p}$ and the pair and triple terms in (2) are divisible by p ,

$$P_{p,n}^{[3]}(u) \equiv (1 + nu)P_{p,n-1}^{[3]}(u) \pmod{(p, u^4)}.$$

Iteration gives (15). The coefficient of u^k in the product is $e_k(1, \dots, m-1)$, which equals the near-diagonal unsigned Stirling number $\begin{bmatrix} m \\ m-k \end{bmatrix}$, proving (16). For the normalized statement we avoid pointwise cancellation of the falling factorial modulo p , which can vanish for special residues of m . Instead reduce the exact polynomial formulas (12)–(14) modulo p ; because 2, 24, and 48 are units for $p \geq 5$, they give exactly the three polynomials in (18). Over $\mathbb{Q}[m]$ these are the normalized identities $e_k(1, \dots, m-1)/(m-1)_{\underline{k}}$, yielding (17) as a polynomial congruence. $\square$

6 The third-column p -adic geometry

Write

$$\Pi_{p,3}(m) = \frac{p+1}{48}F_p(m), \quad (19)$$

where

$$F_p(m) = (p+1)^2m^3 + (p^2-1)m^2 - (34p^2+18p)m + 56p^2 - 8p. \quad (20)$$

For $p \geq 5$, $(p+1)/48$ is a p -adic unit. By Theorem 5.1,

$$F_p(m) \equiv m^2(m-1) \pmod{p}. \quad (21)$$

Thus the shadow has a simple root at 1 and a double root at 0.

6.1 The simple Hensel branch

At $m = 1$,

$$F_p(1) = 24p(p-1), \quad F_p'(1) = 1 - 12p - 29p^2 \equiv 1 \pmod{p}. \quad (22)$$

Theorem 6.1 (Simple Hensel tower). *There is a unique $\alpha_p \in \mathbb{Z}_p$ such that*

$$\alpha_p \equiv 1 \pmod{p}, \quad F_p(\alpha_p) = 0.$$

Moreover,

$$\alpha_p = 1 + 24p - 888p^2 + 58008p^3 - 4715256p^4 + 428583192p^5 + O(p^6). \quad (23)$$

For every integer $m \equiv 1 \pmod{p}$,

$$\nu_p(F_p(m)) = \nu_p(m - \alpha_p). \quad (24)$$

Proof. The congruences in (22) give the existence and uniqueness by Hensel's lemma. Expanding $F_p(1 + \sum_{j \geq 1} a_j p^j) = 0$ recursively gives (23). Factor $F_p(m) = (m - \alpha_p)G_p(m)$ over $\mathbb{Z}_p$. Since $G_p(\alpha_p) = F_p'(\alpha_p)$ is a unit and $m \equiv \alpha_p \equiv 1 \pmod{p}$, $G_p(m)$ remains a unit throughout that residue class, proving (24). $\square$

6.2 Singular extinction at the double shadow root

Let $m = pu$. Direct substitution in (20) gives

$$F_p(pu) = -8p + O(p^2). \quad (25)$$

Theorem 6.2 (Singular extinction). *For every $u \in \mathbb{Z}_p$,*

$$\boxed{\nu_p(F_p(pu)) = 1.} \quad (26)$$

Hence the double root $m \equiv 0 \pmod{p}$ of the Stirling shadow has no lift to a root modulo p^2 :

$$0 \pmod{p} \longrightarrow \emptyset \pmod{p^2}.$$

Proof. Dividing (25) by p gives $F_p(pu)/p \equiv -8 \not\equiv 0 \pmod{p}$ for $p \geq 5$. Thus the valuation is exactly one, uniformly in u . $\square$

This behavior is the opposite of naive branching: the multiplicity-two shadow root disappears from the integral p -adic locus as soon as the exact carry deformation is restored.

6.3 Complete valuation law

For $m \geq 4$, define the scaled third-column amplitude

$$Q_{m,3}^{\text{CCR}} = p^{m-4} \Pi_{p,3}(m). \quad (27)$$

Theorem 6.3 (Complete third-column valuation law). *For every prime $p \geq 5$ and integer $m \geq 4$,*

$$\boxed{\nu_p(Q_{m,3}^{\text{CCR}}) = \begin{cases} m-4, & m \not\equiv 0, 1 \pmod{p}, \\ m-3, & m \equiv 0 \pmod{p}, \\ m-4 + \nu_p(m - \alpha_p), & m \equiv 1 \pmod{p}. \end{cases}} \quad (28)$$

Proof. The prefactor $(p+1)/48$ in (19) is a unit. If $m \not\equiv 0, 1 \pmod{p}$, then (21) is nonzero and $\nu_p(F_p(m)) = 0$. If $m \equiv 0 \pmod{p}$, use Theorem 6.2. If $m \equiv 1 \pmod{p}$, use (24). Finally add the deterministic factor $m-4$ from p^{m-4} . $\square$

7 Discriminant and quadratic ramification

The two roots associated to the extinct residue do not vanish algebraically; they leave $\mathbb{Q}_p$.

Theorem 7.1 (Discriminant and ramified field). *Let F_p be the cubic in (20). Its discriminant is*

$$\boxed{\text{Disc}(F_p) = 4p(p+1)^2 R_p,} \quad (29)$$

where

$$R_p = 9801p^5 + 22672p^4 + 32502p^3 + 14360p^2 - 927p - 8. \quad (30)$$

Thus

$$\nu_p(\text{Disc } F_p) = 1, \quad [\text{Disc } F_p] = [-2p] \in \mathbb{Q}_p^\times / (\mathbb{Q}_p^\times)^2.$$

If $\beta_{p,+}, \beta_{p,-}$ are the two roots distinct from α_p , then

$$\nu_p(\beta_{p,+}) = \nu_p(\beta_{p,-}) = \frac{1}{2} \quad (31)$$

and

$$\boxed{\mathbb{Q}_p(\beta_{p,+}) = \mathbb{Q}_p(\beta_{p,-}) = \mathbb{Q}_p(\sqrt{-2p}).} \quad (32)$$

Proof. A direct cubic discriminant calculation gives (29)–(30). Since $R_p \equiv -8 \pmod{p}$, the discriminant has valuation one. Moreover $R_p/(-8) \equiv 1 \pmod{p}$, hence $R_p/(-8)$ is a square in $\mathbb{Z}_p^\times$ by Hensel's lemma. The factors 4 and $(p+1)^2$ are squares, and $-8 = -2 \cdot 4$, proving the square class $[-2p]$.

Factor over $\mathbb{Z}_p$ as

$$F_p(m) = (m - \alpha_p)Q_p(m),$$

with the leading coefficient absorbed into the quadratic factor. Since $\alpha_p \equiv 1 \pmod{p}$ and $F_p(m) \equiv m^2(m - 1) \pmod{p}$, the quadratic satisfies

$$Q_p(m) \equiv u m^2 \pmod{p}$$

for a unit u . Its constant term is, up to a unit,

$$-F_p(0)/\alpha_p = -8p(7p - 1)/\alpha_p,$$

which has valuation exactly one. After division by its unit leading coefficient, Q_p is therefore Eisenstein. Its two roots have valuation $1/2$ and it is irreducible over $\mathbb{Q}_p$.

For a product of a linear and a quadratic factor, the cubic discriminant differs from the quadratic discriminant by a square resultant factor. Hence the quadratic discriminant has the same square class $[-2p]$. The splitting field of an irreducible quadratic is obtained by adjoining the square root of its discriminant, which proves (32). $\square$

Corollary 7.2 (Ramified square class by $p \pmod{8}$). *If $p \equiv 1, 3 \pmod{8}$, then -2 is a square in $\mathbb{Q}_p$ and*

$$\mathbb{Q}_p(\sqrt{-2p}) = \mathbb{Q}_p(\sqrt{p}).$$

If $p \equiv 5, 7 \pmod{8}$, the field is the other ramified quadratic square class obtained from a nonsquare unit times p .

8 A consolidated main theorem

Theorem 8.1 (CCR-3 arithmetic geometry). *For every prime $p \geq 5$, the third-order chronological carry refinement of Definition 3.1 has the following properties.*

- (a) *Its first three normalized fixed columns are the polynomials (12)–(14).*
- (b) *Modulo p , the entire third jet degenerates to the free product $\prod_{j=1}^n (1 + ju)$, and the normalized fixed columns reduce to the near-diagonal first-kind Stirling polynomials (17).*
- (c) *In the third column the simple shadow root $1 \pmod{p}$ lifts uniquely to $\alpha_p \in \mathbb{Z}_p$, while the double shadow root $0 \pmod{p}$ undergoes singular extinction.*
- (d) *The scaled third column satisfies the complete valuation law (28).*
- (e) *The two nonintegral roots of the exact third-column cubic have valuation $1/2$ and generate $\mathbb{Q}_p(\sqrt{-2p})$.*

Proof. Combine Theorems 4.2, 5.1, 6.1, 6.2, 6.3, and 7.1. $\square$

9 Relation to existing work and novelty boundary

The terminology “carry” is used here for chronological interaction propagation, not for ordinary base- N digit addition. Classical carry processes, especially the Holte matrix, already have deep connections with Eulerian and Stirling structures [4]; a recent analysis makes unsigned Stirling factors explicit in the eigenvectors of the Holte matrix [6]. Accordingly, no claim is made that the appearance of Stirling numbers in a carry-related problem is itself new.

Likewise, p -adic valuations of first-kind Stirling numbers have been extensively studied, including exact and asymptotic valuation phenomena in special families [5, 1]. Hensel lifting, Eisenstein irreducibility, discriminants, Newton polygons, and ramified quadratic extensions are standard p -adic tools. Weighted m -Dyck paths and lower-Hessenberg production matrices also belong to established frameworks [7].

The present contribution is therefore stated at the level of the explicit family proved above. To the best of our knowledge, the specific CCR-3 recurrence, its fixed-column polynomialization, the degeneration to a universal near-diagonal Stirling shadow, and the subsequent resolution of the third-column shadow into a simple Hensel tower together with an extinct integral branch and the exact ramified field $\mathbb{Q}_p(\sqrt{-2p})$ have not previously been isolated as one family-specific theorem chain.

10 Conclusion and open directions

CCR-3 provides a controlled example in which local chronological interactions produce a rigid arithmetic deformation. The free part survives modulo p and becomes a first-kind Stirling structure; the exact interaction terms then decide how the shadow roots lift. In the third column, the answer is asymmetric: the simple root remains integral and produces a Hensel tower, whereas the double root becomes extinct in $\mathbb{Z}_p$ and survives only as a ramified algebraic pair.

The construction is intentionally third-order. A naive continuation of the three-band Hessenberg model should not be interpreted as a complete all-column refinement: at order four, a new subdiagonal/connected interaction can enter for the first time. The natural next problem is therefore not to extrapolate the present recurrence blindly, but to identify and derive the genuine fourth-order connected motif. A second direction is to determine which broader classes of local chronological deformations have Stirling shadows and to classify the possible lifting patterns of multiple shadow roots.

Statements and Declarations

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